

100 SQL Questions Every Fresher Should Know

Categorized SQL Question Bank for Job Interviews

Ye list SQL interview mein sabse zyada poochhe jaane wale 100 questions ko cover karti hai — basics se lekar joins, subqueries, indexes, transactions aur practical scenario-based questions tak. Practice karo aur apni SQL foundation strong banao.

SQL Basics

1. What is SQL and what is it used for?
2. What is the difference between SQL and MySQL?
3. What are the different types of SQL commands (DDL, DML, DCL, TCL, DQL)?
4. What is a database and what is a DBMS?
5. What is the difference between a database and a schema?
6. What are the different data types available in SQL?
7. What is the difference between CHAR and VARCHAR?
8. What is NULL in SQL? How is it different from zero or blank space?
9. What is the difference between DELETE, TRUNCATE and DROP?
10. What is a Primary Key?
11. What is a Foreign Key?
12. What is the difference between Primary Key and Unique Key?

SELECT & Filtering

13. How do you retrieve all columns from a table?
14. How do you retrieve specific columns from a table?
15. What is the use of the WHERE clause?
16. What is the difference between WHERE and HAVING?
17. How do you fetch unique/distinct records from a table?
18. What is the use of the LIKE operator? Give an example.
19. How do you use the BETWEEN operator?
20. How do you use the IN operator?
21. What is the difference between IN and EXISTS?
22. How do you sort query results using ORDER BY?
23. How do you limit the number of rows returned in a query?
24. How do you filter records with NULL values?

Joins

25. What is a JOIN in SQL?
26. What is the difference between INNER JOIN and OUTER JOIN?
27. What is a LEFT JOIN? Give an example.
28. What is a RIGHT JOIN? Give an example.

- 29. What is a FULL OUTER JOIN?
- 30. What is a SELF JOIN? When would you use it?
- 31. What is a CROSS JOIN?
- 32. How is a JOIN different from a Subquery?
- 33. Can you join more than two tables in a single query? How?
- 34. What happens if you don't specify a JOIN condition?
- 35. What is the difference between UNION and UNION ALL?
- 36. How do you find duplicate records using a JOIN?

Aggregate Functions & GROUP BY

- 37. What are aggregate functions in SQL? Name a few.
- 38. What is the difference between COUNT(*) and COUNT(column_name)?
- 39. What is the use of the GROUP BY clause?
- 40. Why can't you use WHERE to filter aggregated results?
- 41. How do you find the second highest salary in a table?
- 42. How do you find duplicate values in a column using GROUP BY?
- 43. What is the difference between SUM() and AVG()?
- 44. How do you count the number of rows in each group?
- 45. What is the order of execution of SQL clauses (SELECT, FROM, WHERE, GROUP BY, HAVING, ORDER BY)?

Subqueries & Set Operations

- 46. What is a subquery? Give an example.
- 47. What is the difference between a subquery and a correlated subquery?
- 48. Can a subquery return multiple rows and columns?
- 49. What is the difference between a subquery in WHERE clause vs FROM clause?
- 50. What is a nested subquery?
- 51. How do you use a subquery with the EXISTS operator?
- 52. What is the INTERSECT operator used for?
- 53. What is the MINUS/EXCEPT operator used for?
- 54. How do you use a subquery to find records above the average value?

Keys, Constraints & Normalization

- 55. What is Normalization? Why is it important?
- 56. What are the different normal forms (1NF, 2NF, 3NF, BCNF)?
- 57. What is Denormalization and when would you use it?
- 58. What is a Candidate Key?
- 59. What is an Alternate Key?
- 60. What is the difference between a Surrogate Key and a Natural Key?
- 61. What is referential integrity?
- 62. What happens when you delete a record referenced by a Foreign Key?

63. What is the ON DELETE CASCADE option used for?

Indexes & Views

- 64.** What is an index in SQL? Why is it used?
- 65.** What is the difference between a Clustered and Non-Clustered Index?
- 66.** Do indexes always improve performance? Why or why not?
- 67.** What is a View in SQL?
- 68.** What is the difference between a Table and a View?
- 69.** Can you update data through a View?
- 70.** What is a Materialized View?
- 71.** What is the difference between a View and a Stored Procedure?
- 72.** How do you drop an existing View?

Functions, Procedures & Triggers

- 73.** What is a Stored Procedure? What are its advantages?
- 74.** What is the difference between a Function and a Stored Procedure?
- 75.** What is a Trigger in SQL? Give an example use case.
- 76.** What are the different types of Triggers (BEFORE, AFTER)?
- 77.** What is a User-Defined Function (UDF)?
- 78.** What is the difference between Scalar and Table-Valued Functions?
- 79.** How do you handle errors in a Stored Procedure?
- 80.** What is a Cursor in SQL? When would you use one?
- 81.** What are the disadvantages of using Cursors?

Transactions & Concurrency

- 82.** What is a Transaction in SQL?
- 83.** What are the ACID properties of a transaction?
- 84.** What is the use of COMMIT and ROLLBACK?
- 85.** What is a SAVEPOINT?
- 86.** What is the difference between COMMIT and SAVEPOINT?
- 87.** What are the different transaction isolation levels?
- 88.** What is a Deadlock? How can it be avoided?
- 89.** What is the difference between Optimistic and Pessimistic Locking?
- 90.** What is the default isolation level in most databases?

Practical & Scenario-Based

- 91.** Write a query to find the Nth highest salary from a table.
- 92.** Write a query to fetch the top 3 records from a table.
- 93.** Write a query to remove duplicate rows from a table.
- 94.** Write a query to swap the values of two columns without using a third variable.
- 95.** Write a query to find employees who earn more than their managers.

96. Write a query to find the total number of employees in each department.
 97. How do you find the current date and time in SQL?
 98. How do you convert a string to a date in SQL?
 99. What is the difference between a Local and Global temporary table?
 100. How would you optimize a slow-running SQL query?
-

Tip: In sabhi questions ko likh kar practice karo, sirf padh kar mat chodo!